8.3: Equivalence Relations

Definition.
Given a partition of a set A, the **relation induced by the partition**, R, is defined on A as follows:

$$\forall x, y \in A,\ x\,R\,y \ \Leftrightarrow \ \begin{array}{l}\text{there is a subset } A_i \text{ of the partition} \\ \text{such that both } x \text{ and } y \text{ are in } A_i.\end{array}$$

Example. Let $A = \{0, 1, 2, 3, 4\}$ and consider the following partition of A:

$$\{0, 3, 4\}, \{1\}, \{2\}.$$

Find the relation R induced by this partition

$0R0$	$1R_1$	$2R_2$	$3R_0$	$4R_0$
$0R_3$			$3R_3$	$4R_3$
$0R_4$			$3R_4$	$4R4$

$\Rightarrow R = \{(0,0), (0,3), (0,4), (1,1), (2,2), (3,0), (3,3), (3,4), (4,0), (4,3), (4,4)\}$

Let A be a set with a partition and let R be the relation induced by the partition. Then R is reflexive, symmetric, and transitive.

Definition.
Let A be a set and R a relation on A, R is an **equivalence relation** if, and only if, R is reflexive, symmetric, and transitive.

Example. Let X be the set of all nonempty subsets of $\{1, 2, 3\}$. Then

$$X = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Define a relation $\mathbf{R}$ on X as follows:

$$\forall A, B \in X,\ A\,\mathbf{R}\,B \Leftrightarrow \text{ the least element of } A \text{ equals the least element of } B.$$

Prove that $\mathbf{R}$ is an equivalence relation on X.

R is reflexive:
The least element of A is also the least element of A, so ARA

R is symmetric:
If ARB, then the least element of A is the least element of B. This means BRA since the least element of B is also the least element of A.

R is transitive:
If ARB, and BRC, then the least element of A is the least element of B, and the least element of B is the least element of C. Since the least element of A and B are the same, then the least element of A is the least element of C, which ARC.

Definition.
Suppose A is a set and R is an equivalence relation on A. For each element a in A, the **equivalence class of** a, denoted $[a]$ and called the **class of** a, is the set of all elements x in A such that x is related to a by R:

$$[a] = \{x \in A | x\, R\, a\}.$$

Example. Let $A = \{0, 1, 2, 3, 4\}$ and define a relation R on A as follows:

$$R = \{\{0,0\}, \{0,4\}, \{1,1\}, \{1,3\}, \{2,2\}, \{3,1\}, \{3,3\}, \{4,0\}, \{4,4\}\}.$$

Find the *distinct* equivalence classes of R.

- $[0] = \{x \in A \mid x R 0\} = \{0, 4\}$
- $[1] = \{x \in A \mid x R 1\} = \{1, 3\}$
- $[2] = \{x \in A \mid x R 2\} = \{2\}$
- $[3] = \{x \in A \mid x R 3\} = \{1, 3\}$
- $[4] = \{x \in A \mid x R 4\} = \{0, 4\}$

Note:
[0]=[4] and [1]=[3]

Distinct equivalence classes: $\{0, 4\}, \{1, 3\}, \{2\}$

Example. Recall the equivalence relation $\mathbf{R}$ defined on the nonempty subsets of $\{1, 2, 3\}$, X:

$$X = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

List the equivalence classes of $\mathbf{R}$.

$[\{1\}] = \{x \in A \mid x \, R \{1\}\} = \{\{1\}, \{1,2\}, \{1,3\}, \{1,2,3\}\}$

$[\{2\}] = \{x \in A \mid x \, R \{2\}\} = \{\{2\}, \{2,3\}\}$

$[\{3\}] = \{x \in A \mid x \, R \{3\}\} = \{\{3\}\}$

Example. Consider the equivalence relation R on a set A:

$$\forall x, y \in A \; x \, R \, y \Leftrightarrow x = y.$$

Describe the distinct equivalence classes of R.

$[a] = \{x \in A \mid x R a\} = \{x \in A \mid x = a\} = \{a\}$

Suppose A is a set, R is an equivalence relation on A and a and b are elements of A.

- If $a \, R \, b$, then $[a] = [b]$.
- Either $[a] \cap [b] = \varnothing$ or $[a] = [b]$.

Example. Let R be the relation of congruence modulo 3 on the set $\mathbb{Z}$:

$$m\,R\,n \Leftrightarrow 3 \mid (m-n).$$

Describe the distinct equivalence classes of R.

$[0] = \{x \in A \mid x R 0\} = \{x \in A \mid x = 3k\} = \{\ldots, -6, -3, 0, 3, 6, \ldots\}$

$[1] = \{x \in A \mid x R 1\} = \{x \in A \mid x = 3k+1\} = \{\ldots, -5, -2, 1, 4, 7, \ldots\}$

$[2] = \{x \in A \mid x R 2\} = \{x \in A \mid x = 3k+2\} = \{\ldots, -4, -1, 2, 5, 8, \ldots\}$

Definition.
Let $m, n \in \mathbb{Z}$ and let $d \in \mathbb{Z}^+$. We say m **is congruent to** n **modulo** d

$$m \equiv n \mod d$$

if, and only if,

$$d \mid (m-n).$$